

Sensitivity Analysis

- A resource allocation problem.

Consider a resource allocation problem and its accompanying optimal simplex tableau

$$\max \quad Z = 10x_1 + 7x_2 + 6x_3 \quad (\text{return in \$})$$

$$\text{s.t.} \quad 3x_1 + 2x_2 + x_3 \leq 36 \quad (\text{Resource 1})$$

$$x_1 + x_2 + 2x_3 \leq 32 \quad (\text{Resource 2})$$

$$2x_1 + x_2 + x_3 \leq 22 \quad (\text{Resource 3})$$

$$x_1, x_2, x_3 \geq 0$$

standard form:

$$\max \quad Z = 10x_1 + 7x_2 + 6x_3$$

$$\text{s.t.} \quad 3x_1 + 2x_2 + x_3 + x_4 = 36$$

$$x_1 + x_2 + 2x_3 + x_5 = 32$$

$$2x_1 + x_2 + x_3 + x_6 = 22$$

$$x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$$

$$\max \quad C^T x$$

$$\text{s.t.} \quad Ax = b$$

$$x \geq 0$$

Optimal tableau:

	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS	
Z	1	3	0	0	1	0	5	146	$\leftarrow Z^*$
x_2	0	1	1	0	1	0	-1	14	
x_5	0	-2	0	0	1	1	-3	2	$\leftarrow x^*$
x_3	0	1	0	1	-1	0	2	8	

$\underbrace{\hspace{10em}}_{B^{-1}}$

Z	x	RHS
1	$C_B^T B^{-1} A - C^T$	$C_B^T B^{-1} b$
$\vec{0}$	$B^{-1} A$	$B^{-1} b$

We consider how changes in data (A, b, c) affect the optimal solution and the optimal value.

- Changes in objective coefficients C .

- A change in C_j for which x_j is nonbasic.

Consider the quantities in the simplex tableau. The only change is $C_j \rightarrow C'_j$

$$\text{Therefore, } \bar{C}'_j = C'_j - C'_B B^{-1} A_j = C'_j - C_B B^{-1} A_j - C_j + C_j - C'_j = \bar{C}_j - (C'_j - C_j).$$

If $\bar{C}'_j \geq 0$, the original optimal solution stays optimal. Otherwise, we use the primal simplex method to find a new optimal solution.

eg Suppose $C_1 = 10$ is changed to $C'_1 = 14$

Since x_1 is nonbasic, $\bar{C}'_1 = \bar{C}_1 - (C'_1 - C_1) = 3 - (14 - 10) = -1$. Therefore we have.

	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	-1	0	0	1	0	5	146
x_2	0	1	1	0	1	0	-1	14
x_5	0	-2	0	0	1	1	-3	2
x_3	0	①	0	1	-1	0	2	8

→

	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	0	0	1	0	0	7	154
x_2	0	0	1	-1	2	0	-3	6
x_5	0	0	0	2	-1	1	1	18
x_1	0	1	0	1	-1	0	2	8

- A change in C_j where x_j is basic.

When C_j is changed to C'_j , C_B is changed to C'_B . We need to recalculate Row 0.

If the current solution is no longer optimal, we use the primal simplex method to find a new optimal solution.

eg Suppose $C_3 = 6$ is changed to $C'_3 = 5$. Then $(C'_B)^T = [C_2 \ C_5 \ C'_3] = [7 \ 0 \ 5]$.

$$\bar{C}'_1 = (C'_B)^T B^{-1} A_1 - C_1 = [7 \ 0 \ 5] \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} - 10 = 2$$

$$\bar{C}'_4 = (C'_B)^T B^{-1} A_4 - C_4 = [7 \ 0 \ 5] \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} - 0 = 2$$

$$\bar{C}'_6 = (C'_B)^T B^{-1} A_6 - C_6 = [7 \ 0 \ 5] \begin{bmatrix} -1 \\ -3 \\ 2 \end{bmatrix} - 0 = 3$$

$$Z' = (C'_B)^T B^{-1} b = [7 \ 0 \ 5] \begin{bmatrix} 14 \\ 2 \\ 8 \end{bmatrix} = 98 + 40 = 138$$

$\bar{C}_2, \bar{C}_3, \bar{C}_5$ stay 0 because they are the reduced costs of basic variables.

	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	2	0	0	2	0	3	138
x_2	0	1	1	0	1	0	-1	14
x_5	0	-2	0	0	1	1	-3	2
x_3	0	1	0	1	-1	0	2	8

Remark: If $C_3 = 6$ is changed to $C'_3 = 3$, then the current solution is no longer optimal.

- Changes in the limitation of Resource b.

Suppose that b_i is changed to b'_i . Consider the simplex tableau. The last column needs to be updated.

$$B^{-1}b', \quad C_0^T B^{-1}b'$$

If the current solution is no longer feasible, we use the dual simplex method to find a new optimal solution.

e.g. Suppose $b_3 = 22 \rightarrow b'_3 = 24$ Then

$$B^{-1}b' = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -3 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 36 \\ 32 \\ 24 \end{bmatrix} = \begin{bmatrix} 12 \\ -4 \\ 12 \end{bmatrix}$$

$$C_0^T B^{-1}b' = [7 \ 0 \ 6] \begin{bmatrix} 12 \\ -4 \\ 12 \end{bmatrix} = 156$$

Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS	
Z	1	3	0	0	1	0	5	156
x_2	0	1	1	0	1	0	-1	12
x_5	0	-2	0	0	1	1	-3	-4
x_3	0	1	0	1	-1	0	2	12

→

	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	0	0	0	$\frac{5}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	150
x_2	0	0	1	0	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	10
x_5	0	1	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{3}{2}$	2
x_3	0	0	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	10

• Changes in technical coefficient a_{ij}

- If the change is associated with a nonbasic variable, i.e., x_j is nonbasic, then only the column of x_j needs to be updated.

$$B^{-1}A'_j, \quad \bar{C}_j' = C_B^T B^{-1}A'_j - C_j$$

Again, if the current solution is no longer optimal, we use the primal simplex method to find a new optimal solution.

eg Suppose $a_{31} = 2 \rightarrow a'_{31} = 1$, that is, $A_1 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} \rightarrow A'_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$. Then

$$B^{-1}A'_1 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -3 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$$\bar{C}_1' = C_B^T B^{-1}A'_1 - C_1 = \begin{bmatrix} 7 & 0 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} - 10 = -2$$

	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	-2	0	0	1	0	5	146
x_2	0	2	1	0	1	0	-1	14
x_5	0	1	0	0	1	1	-3	2
x_3	0	-1	0	1	-1	0	2	8

→

	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	0	0	0	3	2	-1	150
x_2	0	0	1	0	-1	-2	5	10
x_1	0	1	0	0	1	1	-3	2
x_3	0	0	0	1	0	1	-1	10

→

	Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	0	0	0	$\frac{14}{5}$	$\frac{8}{5}$	0	152
x_6	0	0	$\frac{1}{5}$	0	$-\frac{1}{5}$	$-\frac{2}{5}$	1	2
x_1	0	1	$\frac{2}{5}$	0	$\frac{2}{5}$	$-\frac{1}{5}$	0	8
x_3	0	0	$\frac{1}{5}$	1	$-\frac{1}{5}$	$\frac{3}{5}$	0	12

- If the change is associated with a basic variable, i.e., x_j is basic, then every block in the simplex tableau will be changed. Then we have to resolve the problem from beginning.

• Addition of a new Variable

When adding a new variable, a new column will be added to the current simplex tableau.

If the reduced cost of the new variable is negative, we use the primal simplex method to find an optimal solution.

e.g.: A new variable x_7 is added along with technical coefficients A_7 and an objective coefficient C_7 .

Consider

$$\max \quad Z = 10x_1 + 7x_2 + 6x_3 + 8x_7$$

$$\text{s.t.} \quad 3x_1 + 2x_2 + x_3 + x_4 + 2x_7 = 36$$

$$x_1 + x_2 + 2x_3 + x_5 + 3x_7 = 32$$

$$2x_1 + x_2 + x_3 + x_6 + x_7 = 22$$

$$x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0.$$

That is, $A_7 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ and $C_7 = 8$. Then

$$B^{-1}A_7 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -3 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \bar{C}_7 = C_7 - B^{-1}A_7^T C = [7 \ 0 \ 6] \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - 8 = -1$$

	Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
Z	1	3	0	0	1	0	5	-1	146
x_2	0	1	1	0	1	0	-1	1	14
x_5	0	-2	0	0	1	1	-3	2	2
x_3	0	1	0	1	-1	0	2	0	8



	Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
Z	1	2	0	0	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{7}{2}$	0	147
x_2	0	2	1	0	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	0	13
x_7	0	-1	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{3}{2}$	1	1
x_3	0	1	0	1	-1	0	2	0	8

• Addition of a new constraint

- If the current optimal solution satisfies the new constraint, then the current optimal solution stays optimal.
- Otherwise, we first add the constraint in the simplex tableau with a new slack variable, then zero out its coefficients of the basic variables.

e.g.: Suppose that we add $x_1 + x_2 + x_3 \leq 20$ to the original problem. Then first we have a simplex tableau with one more row and one more column:

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	3	0	0	1	0	5	0	146
x_2	0	1	1	0	1	0	-1	0	14
x_5	0	-2	0	0	1	1	-3	0	2
x_3	0	1	0	1	-1	0	2	0	8
x_7	0	1	1	1	0	0	0	1	20

new slack variable

canonical structure ruined

new constraint

Then we zero out the coefficients of x_2 and x_3 by pivoting. That is, $R_4 - R_1 - R_3 \rightarrow R_4$.

The resulting tableau will be infeasible. We can use the dual simplex method to solve it.

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	3	0	0	1	0	5	0	146
x_2	0	1	1	0	1	0	-1	0	14
x_5	0	-2	0	0	1	1	-3	0	2
x_3	0	1	0	1	-1	0	2	0	8
x_7	0	-1	0	0	0	0	-1	1	-2

→

	z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
z	1	0	0	0	1	0	2	3	140
x_2	0	0	1	0	1	0	-2	1	12
x_5	0	0	0	0	1	1	-1	-2	6
x_3	0	0	0	1	-1	0	1	1	6
x_1	0	1	0	0	0	0	1	-1	2